1 SET THEORY

INTRODUCTION :

☼ **Set :**

A collection or an aggregate of well defined objects is called a set. These objects which belong to the set are called elements or the members of the set.

Sets are usually denoted by capital letters A, B, C etc. and their elements by small letters a, b, c etc.

☼ **Notations :**

If a is an element of a set A, we write $a \in A$.

If b is not an element of set B, we write $b \notin B$.

☼ **Representation of a set :**

A set can be represented in two ways :

(i) The list method

(ii) The property method.

In the list method, we actually list the elements belonging to the set.

In the property method, the characteristics of all the elements of the set is stated.

Examples : $A = \{x / x \in N, 1 < x < 5\}$

$= \{2, 3, 4\}$

$B = \{x / x \in N, x^4 = 81\}$

$= \{3\}$

$C = \{x / x \in N, x^2 < 10\}$

$= \{1, 2, 3\}$

$D = \{x / x \in Z, x^2 \leq 25\}$

$= \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$

$$E = \{x / x^3 - 3x^2 - 4x = 0\}$$
$$= \{-1, 0, 4\}$$

$\because x^3 - 3x^2 - 4x = 0$

$\therefore x(x^2 - 3x - 4) = 0$

$\therefore x(x - 4)(x + 1) = 0$

$\therefore x = 0, x = 4, x = -1$

☼ Finite sets and Infinite sets :

Sets containing finite number of elements are called finite sets.

Set which is not finite is called an infinite set.

☼ Empty set :

A set which contains no element at all is called an empty set. It is denoted by { } or ϕ.

☼ Singleton set :

A set consisting of a single element is called a singlet set.

☼ Subset :

If each element of a set A is also an element of a set B, then A is called a subset of B.

B is called the super set of A. Symbolically we write $A \subseteq B$.

Note : (i) Every set is a subset of itself.

(ii) Null set is a subset of every set.

☼ Proper subset of a set :

The set A is called the proper subset of B if and only if each element of A is the element of B and there is at least one element of B which is not an element of the set A.

i.e. A is a proper subset of B if $A \subseteq B$ and $A \neq B$.

If A is a proper subset of B, we write $A \subset B$.

The set itself and null set are called improper subsets of the set.

☼ Family of sets :

If we have a set whose elements are themselves sets, then the set formed is called the family of sets.

☼ Power set :

The family of all the subsets of a given set is called the power set. Power set of a set A is denoted by $P(A)$.

Note : (i) ϕ and A are both elements of $P(A)$.

(ii) If $A = \phi$, then $P(A) = \{\phi\}$.

(iii) If $A = \{a\}$ then $P(A) = \{\phi, \{a\}\}$.

If a finite set A has n elements, then the power set of A has 2^n elements.

OR

The number of subsets of a finite set A with n elements is given by 2^n.

☼ Universal set (U) :

Generally while discussing a mathematical problem, we think of a definite set such that all sets under discussion are its subsets. This definite set is called the universal set with reference to that problem.

☼ Compliment of a set with respect to the universal set :

Let U be the universal set and A be its subset. The complement of A is the set of all those elements of U which are not elements of A.

Symbolically, compliment of a set A is written as A'.